

Passages - What is Life?

Cells are the smallest entities that have the core characteristics of life. This is the basis of what biologists call cell theory: to the best of our knowledge, everything that is alive on the planet is either a cell or made from a collection of cells. The cell is the simplest thing that can be said, definitively, to be alive.

Bacteria and other sorts of microbial cells ('microbe' is a general term for all microscopic organisms that can live as single cells) are by far the most numerous life forms on Earth. They inhabit every environment, from the high atmosphere to the depths of the Earth's crust.

Rudolf Virchow, wrote in 1858 'that every animal appears as a sum of vital units, each of which bears in itself the complete characteristics of life'.

This means that all living organisms, regardless of their size or complexity, emerge from a single cell.

Quite a lot of what we know about the uncontrolled cell divisions of cancer cells came first from studying the humble yeasts.

Chromosomes are the physical structures present in all cells that contain the genes.

Apart from a few specialized exceptions - like red blood cells which, as they mature, lose their entire nucleus and therefore all their genes - every cell in your body contains a copy of your entire complement of genes.

Every time a cell divides, the entire set of

genes must be copied and shared equally between the two newly formed cells. This means that cell division is the fundamental example of reproduction in biology.

Each of your chromosomes has at its core a single, unbroken molecule of DNA. These can be extremely long and each can contain hundreds or even thousands of genes arranged in a chain, one after another.

The four-letter alphabet of DNA is arranged into three-letter 'words' along each strand of the DNA ladder, with most of those short words corresponding to one specific amino acid building block of a protein. You can think of a gene as being the sequence of DNA words needed to make a specific protein. For example, a human gene with the name beta-globin contains its essential information in 441 DNA 'letters' (that is, nucleotide bases), which spell out 147 three-letter DNA 'words', which the cell translates into a protein molecule that is 147 amino acids long. In this case, the beta-globin protein helps form the oxygen-carrying pigment called haemoglobin, found in red blood cells, that keeps your body alive and makes your blood look red.

The ability to understand the genetic code solved a key mystery at the heart of biology. It showed how the static instructions stored in the genes could be turned into the active protein molecules that build and operate living cells. Breaking this code paved the way to today's world where biologists can readily describe, interpret and modify gene sequences.

Genomes are the complete set of genes or genetic material present in a cell or organism

The copying of the genes and the division of the cell must, therefore, be closely co-ordinated. This co-ordination is achieved by the cell cycle, the process that orchestrates the birth of every new cell.

Mutations occur when the DNA sequence of a gene has been altered, rearranged or deleted. This is usually either the result of damage inflicted on the cell - by UV radiation or chemical damage, for example - or due to the occasional mistakes that can occur during the processes of DNA replication and cell division. The cell has sophisticated mechanisms to spot and repair most of these errors, which means that mutations tend to be rather rare.

The mutations that differentiate your inborn characteristics and mine, and both of us from our prehistoric ancestors, add up to a tiny fraction - less than one percent - of your total complement of DNA code.

Natural selection is the intensely creative process that has produced us. Without ever deviating from the laws of science or invoking supernatural phenomena, evolution by natural selection has generated populations of increasingly complex and diverse creatures over aeons of time.

The wild mustard plant gave rise to broccoli, cabbage, cauliflower, kale and kohlrabi.

For evolution by natural selection to take

place, living organisms must have three crucial characteristics: ability to reproduce, a hereditary system, and the possibility of variability (also death)

In complex eukaryote organisms, further variability is introduced during the process of sexual reproduction, when parts of chromosomes are reshuffled during the cell divisions that produce sex cells. That is the main reason siblings are genetically different from each other: if their parents' genes are like a deck of cards, they are each dealt a different genetic 'hand'

Variation by exchanging DNA sequences directly, between different individuals is called horizontal gene transfer. It is one of the reasons the genes that make certain bacteria resistant to antibiotics can spread rapidly through whole populations of bacteria, and even from one unrelated species to another. This transfer also makes it harder to trace some lineages back through evolutionary time

All the different life forms that share our planet are our relatives.

He argued that chemical reactions were not just an interesting feature of cellular life - they were one of life's defining features. Pasteur summarized this brilliantly when he said that 'chemical reactions are an expression of the life of the cell'. The vast array of chemical reactions occurring in living organisms is called metabolism (the chemistry of life)

Enzymes are catalysts: that means they facilitate and speed up chemical reactions, often dramatically.

Life uses twenty different amino acids.

Each of these amino acids have side molecules, branching off from the main polymer chain, that grant them distinct chemical properties. For example, some amino acids have positive or negative charges, others are either attracted to or repelled by water, and some are capable of easily forming bonds with other molecules. By stringing together different combinations of amino acids, each with different side molecules, cells can create a huge range of different protein polymer molecules.

The outcome of all of this protein diversity is a maelstrom of chemical reactions being carried out in every cell at all times. If you could imagine looking inside a living cell with eyes that could perceive the molecular world, your senses would be assaulted by a boiling tumult of chemical activities.

Each enzyme protein molecule provides enclosed spaces and docking sites which have specific shapes that are tailor-made, at the scale of individual atoms, for associating with the specific molecules that they work with.

All these different chemistries must be carried out both simultaneously and very close together in the narrow confines of the cell. This is only possible because the various enzymes do not each require different extreme temperatures, pressures or acid or alkaline conditions, such as those found in industrial chemical facilities. If they did, they would not all be able to co-exist in such close proximity. Many of these metabolic reactions still need to be kept separate from one another, however. They must not interrupt each other, and all their specific chemical

requirements must be met. Key to answering this challenge is compartmentation.

There is good evidence that gradients of signalling molecules across the bodies of developing organisms do indeed contribute to the appearance of sophisticated biological forms.

Alan Turing devised a set of mathematical equations that predicted the behaviour of chemical substances interacting with each other, and so undergoing specific chemical reactions as they diffuse through a structure. Unexpectedly, his equations, which he called reaction-diffusion models, could arrange chemical substances into elaborate and often rather beautiful spatial patterns. By tweaking the parameters of his equations, the two substances could organize themselves into evenly spaced spots, stripes or blotches, for example. The attractive thing about Turing's model is that the patterns emerge spontaneously.

'Mathematics is the art of the perfect. Physics is the art of the optimal. Biology, because of evolution, is the art of the satisfactory.'

The life forms that survive natural selection persist because they work, not necessarily because they do things in the most efficient or straightforward way possible.

Information theory is a way to move from describing complexity to understanding complexity.

Cancer is an inevitable result of the cell's capacity to adapt and evolve

To feed applied science by starving basic science is like economizing on the foundations of a building so that it may be built higher. It is only a matter of time before the whole edifice crumbles.'

Ultimately, life emerges from the relatively simple and well-understood rules of chemical attraction and repulsion, and the making and breaking of molecular bonds.

The critical fact about DNA is that each gene is a linear sequence of information written in the four-letter language of DNA. Linear sequences are a familiar and highly effective strategy for storing and conveying information

Ultimately, the cells in each of your organs are different because they use very different combinations of genes. In fact, only about 4,000, or a fifth, of your total set of genes are thought to be turned on and used by all the different types of cells in your body to support the basic operations needed for their survival. The rest are only used sporadically, either because they perform specific functions only required by some types of cell, or because they are only needed at specific times.

The positive feedback loop, which can form irreversible switches that once turned on are never turned off (ripening of apple, ethylene gas)

Mechanisms that produce switches that can flip reversibly between 'on' and 'off' states, or oscillators that rhythmically and continuously pulse 'on' and 'off' (differentiating between night and day).

The overriding ambition should be to capture meaning, rather than simply catalogue complexity.

Signalling pathways transmit information within cells, between cells, between organs, between whole organisms, between populations of organisms and even between different species across whole ecosystems.